Anchored Data at Four Epochs

Experiment index

Question and Context

Would substantially longer training improve the two-epoch reference, or simply strengthen the wrong decision patterns?

Controlled Change

Train a fresh adapter for four total epochs on the same anchored data. This was not continuing from the saved two-epoch adapter.

- Fresh base Qwen/Qwen2.5-1.5B-Instruct plus fresh LoRA; not a resumed adapter.
- LoRA rank 8, alpha 16, q_proj/v_proj, dropout 0.05, bias none.
- Learning rate 2e-4, cosine schedule, warmup fraction 0.03, seed 1.
- Owner overrides: context 1,024, train/eval batch one, accumulation 16, raw caps 1,000/200.
- Assistant-only SFT token cross-entropy; prompt/padding tokens do not provide target loss.
- Recorded training/inference system prompt hash a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.

The usable training/validation counts stayed 974/195, with the same broad class counts and no exact duplicates. The learning-rate schedule now spans four epochs.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / four epochs	150	37/150 (0.247)	0.440	0.628	0.263	0.103
Anchored / four epochs	30	4/30 (0.133)	0.400	0.528	0.300	0.050

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / four epochs (150 cases)	0.372	0.087
Anchored / four epochs (30 cases)	0.348	0.200

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / four epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	39	36	1	5
CLARIFY	14	19	4	3
REFUSE	17	3	4	5

Decision Matrix: Anchored / four epochs (30 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	2	7	1	0
CLARIFY	1	8	1	0
REFUSE	3	0	1	6

Result Provenance

- Anchored / four epochs: 2026-10-07 UTC, 150 cases.
- Anchored / four epochs: 2026-10-07 UTC, 30 cases.

Interpretation

Compared with two epochs, combined task successes fell from 44 to 37, macro-F1 from 0.514 to 0.372, and invalid outputs increased from two to thirteen.

Correct CLARIFY routing rose from 11/40 to 19/40, but correct ACT fell from 57/81 to 39/81 and correct REFUSE from 16/29 to 4/29. Higher clarification wording scores therefore concealed a harmful routing tradeoff.

Shift success also fell, from five to four, and invalid outputs rose from two to six. Its macro-F1 was nearly unchanged, but the class-specific behavior changed substantially.

Why Lower Loss Was Not Enough

Validation token loss fell to 0.2364 from the one-epoch run's 0.3107. That objective rewards the supplied answer tokens; it does not directly optimize routing macro-F1, execution success or semantic explanations.

The loss reduction does not prove useful generalization, nor does the regression prove memorization. It does show that more epochs are not a reliable fix under this recipe.

Decision and Limits

Keep two epochs as the reference. The saved audit separates wrong decisions, wording-score mismatch, argument errors and invalid output without repairing results.

This was one seed, and changing epochs changes the schedule horizon. Do not generalize it to every learning rate, dataset or local model.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs4.json
- anchored-epochs2-vs4-failure-audit.md
- anchored-epochs2-vs4-failure-audit.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.